12.1.1 Maximum-Variance Formulation of PCA

Pattern Recognition and Machine Learning, C.M.Bishop

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2. Problem Setup
3. Maximum-Variance Formulation

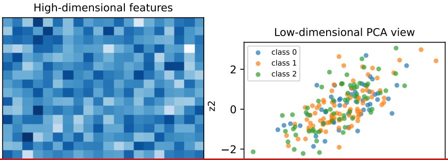
Motivation

Why do we need PCA?

- Modern data sets often live in a high-dimensional ambient space $\mathbb{R}^D$ (images, text features, genomic data, ...).
- Directly working in this space is expensive and can hide the essential structure of the data.
- Principal Component Analysis (PCA) is a linear dimensionality reduction method that searches for directions along which the data vary the most.

Focus of this talk

- We follow Section 12.1.1 of Pattern Recognition and Machine Learning (Bishop, 2006).
- We explain the maximum-variance formulation of PCA in an intuitive but rigorous way.
- The goal is that, even as beginners, you can understand both the geometry and the optimization behind PCA.



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Data and Linear Projection

Data matrix and centering

- We observe N data points $\mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^D$.
- We assume the data have been centered:

$$\frac{1}{N} \sum_{n=1}^{N} \mathbf{x}_n = \mathbf{0}.$$

- Centering simplifies the analysis and makes the origin a natural reference point.

Linear projection onto one direction

- We first look for one direction $\mathbf{u}_1 \in \mathbb{R}^D$ onto which we project the data.
- For each data point $\mathbf{x}_n$, the scalar projection is

$$z_{n1} = \mathbf{u}_1^\top \mathbf{x}_n.$$

- We constrain $\mathbf{u}_1$ to have unit length, $\|\mathbf{u}_1\|^2 = \mathbf{u}_1^\top \mathbf{u}_1 = 1$, so that the scale of the projection is well-defined.

Sample Covariance and Projected Variance

Sample covariance matrix

- The (empirical) covariance matrix of the centered data is

$$\mathbf{S} \stackrel{\text{def}}{=} \frac{1}{N}\sum_{n=1}^{N} \mathbf{x}_n \mathbf{x}_n^\top \in \mathbb{R}^{D\times D}.$$

- $\mathbf{S}$ captures how each pair of coordinates varies together.

Variance along a direction

- The sample mean of the projections is zero:

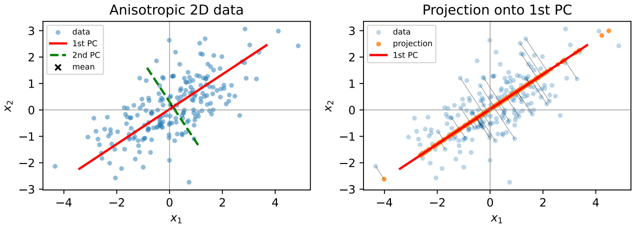
$$\frac{1}{N}\sum_{n=1}^{N} z_{n1} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{u}_1^\top \mathbf{x}_n = \mathbf{u}_1^\top \left(\frac{1}{N}\sum_{n=1}^{N} \mathbf{x}_n \right) = 0.$$

- The sample variance of the projected data is

$$\frac{1}{N}\sum_{n=1}^{N} z_{n1}^2 = \frac{1}{N}\sum_{n=1}^{N} \left(\mathbf{u}_1^\top \mathbf{x}_n\right)^2 = \mathbf{u}_1^\top \mathbf{S}\, \mathbf{u}_1.$$

- Thus, the variance along direction $\mathbf{u}_1$ can be written compactly as a quadratic form in $\mathbf{u}_1$.

2D Toy Example of PCA



Interpretation

- **Left panel:** Anisotropic 2D data cloud together with the first (red) and second (green) principal axes and the sample mean (black cross).
- **Right panel:** The same data points, together with their orthogonal projections onto the first principal component (red line).
- Points are moved along short perpendicular segments to the red line; the coordinates on this line are exactly the one-dimensional PCA representation.

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Optimization Problem for the First Component

Principle

- PCA chooses directions that maximize the variance of the projected data.
- The first principal component $\mathbf{u}_1$ is the unit vector along which the projected data have the largest variance.

Constrained optimization problem

- Using the expression for the variance, we want to solve

$$\max_{\mathbf{u}_1} \quad \mathbf{u}_1^\top \mathbf{S}\,\mathbf{u}_1 \quad \text{subject to} \quad \mathbf{u}_1^\top \mathbf{u}_1 = 1.$$

- This is a quadratic objective with a quadratic (sphere) constraint.
- We can solve it using the method of Lagrange multipliers.

Lagrangian and Eigenvalue Equation

Lagrangian

- Introduce a Lagrange multiplier λ_1 and define

$$\mathcal{L}(\mathbf{u}_1, \lambda_1) = \mathbf{u}_1^\top \mathbf{S}\, \mathbf{u}_1 - \lambda_1(\mathbf{u}_1^\top \mathbf{u}_1 - 1).$$

Stationary condition

- Taking the derivative with respect to $\mathbf{u}_1$ and setting it to zero gives

$$\frac{\partial \mathcal{L}}{\partial \mathbf{u}_1} = 2\mathbf{S}\, \mathbf{u}_1 - 2\lambda_1 \mathbf{u}_1 = \mathbf{0}.$$

- Hence, any stationary point satisfies

$$\mathbf{S}\, \mathbf{u}_1 = \lambda_1 \mathbf{u}_1,$$

which is the eigenvalue equation for the covariance matrix $\mathbf{S}$.

Choosing the largest eigenvalue

- One can show that the objective value at a stationary point is equal to the corresponding eigenvalue λ_1.
- Therefore, the maximum variance is obtained by choosing $\mathbf{u}_1$ to be the eigenvector of $\mathbf{S}$ associated with its largest eigenvalue.

Subsequent Principal Components

Adding a second component

- After finding $\mathbf{u}_1$, we look for a second direction $\mathbf{u}_2$.
- We again maximize the variance of the projection

$$z_{n2} = \mathbf{u}_2^\top \mathbf{x}_n,$$

  subject to $\|\mathbf{u}_2\|^2 = 1$.
- In addition, we require $\mathbf{u}_2$ to be orthogonal to $\mathbf{u}_1$:

$$\mathbf{u}_1^\top \mathbf{u}_2 = 0.$$

Result

- Solving this constrained problem again leads to an eigenvalue equation for $\mathbf{S}$.
- The optimal $\mathbf{u}_2$ is the eigenvector corresponding to the second largest eigenvalue of $\mathbf{S}$.
- Repeating this reasoning, we obtain a sequence of orthonormal eigenvectors

$$\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_M,$$

  which form the first M principal components.

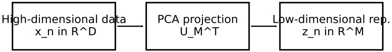
Low-Dimensional Representation and Geometry

Low-dimensional coordinates

- Collect the first M principal components into a matrix
$$\mathbf{U}_M = [\mathbf{u}_1, \ldots, \mathbf{u}_M] \in \mathbb{R}^{D \times M}.$$
- The M-dimensional representation of $\mathbf{x}_n$ is
$$\mathbf{z}_n = \mathbf{U}_M^\top \mathbf{x}_n \in \mathbb{R}^M.$$
- These coordinates keep as much variance of the original data as possible among all linear M-dimensional projections.

Geometric interpretation

- If the data cloud is roughly ellipsoidal, the eigenvectors of $\mathbf{S}$ give the principal axes of this ellipsoid.
- PCA aligns the coordinate system with these axes so that most of the spread lies in the first few coordinates.
- This is why PCA is often used for visualization (e.g., $M = 2$) and as a preprocessing step for other learning algorithms.



Summary of Section 12.1.1

(i) Start from centered data $\mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^D$ and compute the sample covariance matrix

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{x}_n \mathbf{x}_n^\top .$$

(ii) The variance of the projection along a unit vector $\mathbf{u}$ is $\mathbf{u}^\top \mathbf{S}\, \mathbf{u}$.

(iii) Maximizing this variance under the unit-norm constraint leads, via Lagrange multipliers, to the eigenvalue equation

$$\mathbf{S}\, \mathbf{u} = \lambda \mathbf{u}.$$

(iv) The principal components are the eigenvectors of $\mathbf{S}$, ordered by decreasing eigenvalues; they define directions of maximal variance in the data.

(v) Projecting onto the first M principal components yields an M-dimensional representation that preserves as much variance as possible among all linear projections.

References I

Bishop, C. M. (2006). Pattern recognition and machine learning. New York: Springer.